

Experiment-25

Aim

To perform image sharpening using gradient masking and enhance the edges and fine details of the input image. .

Procedure

1. Select and load the input image.
2. Convert the image into grayscale.
3. Define the two gradient masks G_x and G_y
4. Apply the masks to the image using convolution.
5. Calculate the gradient magnitude from the two gradient images.
6. Add the gradient information to the original image.
7. Clip the pixel values to the range 0–255.
8. Display the original, gradient, and sharpened images.
9. Observe the enhancement of edges and fine details.

Python Code

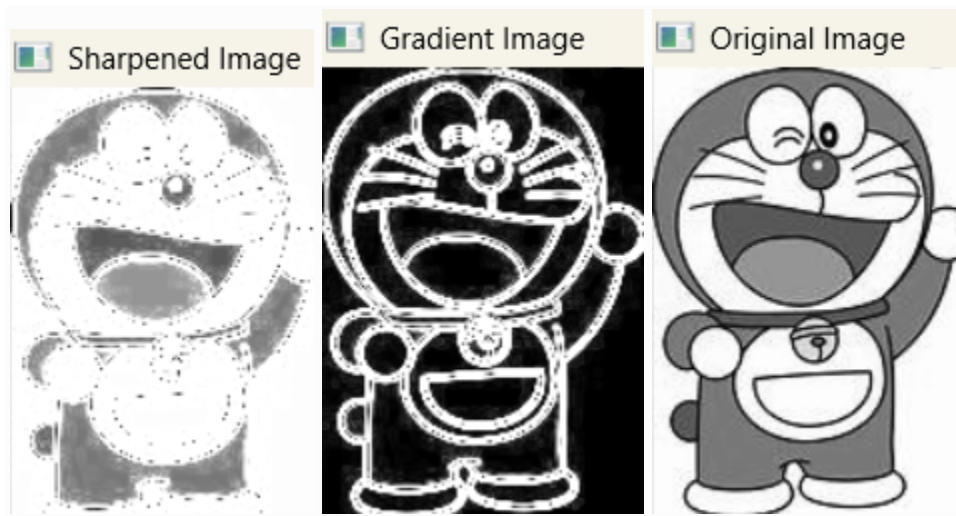
```
import cv2
import numpy as np
img = cv2.imread("image.jpg")
if img is None:
    print("Error: Image not found")
    exit()
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
Gx = np.array([
    [-1, -2, -1],
    [ 0,  0,  0],
    [ 1,  2,  1]
], dtype=np.float32)
Gy = np.array([
    [-1,  0,  1],
```

```
[-2, 0, 2],  
[-1, 0, 1]  
, dtype=np.float32)  
gradient_x = cv2.filter2D(gray, cv2.CV_32F, Gx)  
gradient_y = cv2.filter2D(gray, cv2.CV_32F, Gy)  
gradient = cv2.magnitude(gradient_x, gradient_y)  
gradient = cv2.convertScaleAbs(gradient)  
sharpened = cv2.add(gray, gradient)  
cv2.imshow("Original Image", gray)  
cv2.imshow("Gradient Image", gradient)  
cv2.imshow("Sharpened Image", sharpened)  
cv2.waitKey(0)  
cv2.destroyAllWindows()
```

Sample Input:



Output:



Results:

The image was successfully sharpened using gradient masking. The Sobel gradient masks detected the intensity changes and edges in the image, and the gradient information was added to the original image. As a result, the edges and fine details were enhanced, producing a sharper image.